Martin Satrio, Ph.D.

martin.satrio@gmail.com

https://martinsatrio.github.io

Professional Experience

Sep 2025–· · · · **Meteorological Radar Scientist**, Rainmaker Technology Corporation, Norman, OK

- Designed and built Rainmaker's radar validation workbench from the ground up (Python/Flask, JavaScript, Py-ART, xarray): analysts pull NEXRAD Level II by site and time window, apply dual-pol QC and beam-blockage correction, overlay ASOS/SNOTEL, drone telemetry, and soundings, outline seeding signatures, and estimate precipitation with a 27-member, HCA-aware Z–R ensemble. Maintained through versioned releases and automated Python and JavaScript test suites.
- Lead radar analyst for the 2025–2026 Pendleton, OR program, identifying 55 seeding signatures across 52 IOPs and writing 40+ validation reports that quantify precipitation produced and identify the atmospheric and operational variables behind seeding outcomes.
- Leading an automated validation-report pipeline that fuses MRMS (GRIB2), GOES-18 cloud-top products, HRRR/GFS fields, ceilometer, sounding, and drone data with flight logs into a draft IOP report for analyst review, across multiple program areas.
- Co-developing an automated seeding-signature detector (tobac); verified it against analyst-labeled radar cases with object-based statistics and a chronological holdout, raising POD from 7% to 52% while cutting false-alarm ratio from 78% to 41% during calibration.
- Built a 3-D mission replay app (Three.js, Python, Docker) that animates complete operations (NEXRAD volumes, signature polygons, drone and balloon tracks, ensemble QPE, terrain) for 16 cases, giving clients and non-specialists an intuitive view of operations; now being extended to prospective operations in new domains.
- First author of a season-wide study of 71 radar-observed precipitation enhancements across all program areas (*Atmosphere*, in revision), including drone-vs-radiosonde comparisons, IOP-clustered bootstrap confidence intervals, and QPE sensitivity tests.
- Work with business development and brand/comms to turn scientific results into client-facing figures, validation summaries, and proposal material; presented to internal stakeholders, state water-resource agencies, and academic conferences.
- Contributing to in-house NWP and cloud modeling (WRF, CM1) on Modal and AWS; use GFS and HRRR guidance in the forecasting workflow and to characterize seeding environments.

Professional Experience (continued)

2023–2025 **Post-Doctoral Research Associate**, Cooperative Institute for Severe and High-Impact Weather Research & Operations (CIWRO), Norman, OK

- Led a project with two graduate students quantifying how backing winds and effective inflow-layer depth influence supercell evolution and tornado-like vortex development in CM1 simulations.
- Integrated multi-Doppler wind/reflectivity syntheses into a diabatic Lagrangian framework to retrieve thermodynamic structure in observed supercells.
- Improved Python and FORTRAN pipelines for radar quality control, objective analysis, and multi-Doppler synthesis, increasing reproducibility and reducing manual processing.
- Compiled and analyzed ground-based, airborne, and WSR-88D observations from TORUS to evaluate internal boundaries and updraft characteristics.
- Collaborated with the Warn-on-Forecast (WoFS) team to assimilate quality-controlled radar observations from the 22 May 2023 field campaign.
- Participated in DELTA (2024) and LIFT (2024 & 2025) field campaigns collecting low-level flow observations via mobile mesonet, windsonde, and LiDAR platforms.

2019–2023 **Graduate Research Assistant**, CIWRO & OU School of Meteorology, Norman, OK

- Quantified low-level supercell dynamics using multi-Doppler and diabatic Lagrangian analyses of the 17 May 2019 TORUS case.
- Co-led a project on dissipation of the 10 December 2021 long-track tornado by comparing tree-fall observations with large-eddy simulated vortex behavior over analogous terrain.
- Analyzed WSR-88D cell-merger behavior and its relationship to mesocyclone evolution, contributing to a peer-reviewed Monthly Weather Review publication.
- Participated in PERiLS, TORUS, and TORUS-LiTE collaborative field campaigns collecting supercell observations across the United States.

2017–2019 **Graduate Research Assistant**, Advanced Radar Research Center & OU School of Meteorology, Norman, OK

- Enhanced FORTRAN vortex-model code to test sensitivity to translational velocity, surface roughness length, and terrain structure.
- Led quantitative analyses of terrain effects on tornado dynamics, resulting in a first-authored publication in the Journal of Atmospheric Sciences.
- Coordinated field experiments using the Atmospheric Imaging Radar to sample supercell structures across the Great Plains.

2015–2024 **Teaching Experience / Mentoring**, OU School of Meteorology, Norman, OK

- Lead instructor for Atmospheric Dynamics III (Fall 2023 and Fall 2024), including course design, lectures, and assessments for a mathematically intensive senior-level course.
- Mentored senior capstone teams (Fall 2022 and Fall 2023), guiding research questions, Python-based analysis workflows, and technical presentation development.
- Served as teaching assistant across dynamics, programming, and introductory meteorology courses (2015–2022), with regular office hours, grading, and occasional lecturing.
- Tutored University of Oklahoma student-athletes in mathematics (Fall 2015).

Education

2019–2023 **Ph.D. Meteorology**, University of Oklahoma
Dissertation: *A Kinematic and Thermodynamic Analysis of the 17 May 2019 McCook / Farnam, Nebraska Tornadic Supercell*
General Exam: *A Proposed Observation Network to Investigate Deep Convection Initiation and Early Evolution of Supercells in the Great Plains and Southeast*
Advisors: *Dr. Michael Coniglio and Dr. Erik Rasmussen*

2017–2019 **M.S. Meteorology**, University of Oklahoma
Thesis: *The Effects of Terrain on an Idealized Tornado using a Large-Eddy Simulation*
Advisors: *Dr. David Bodine and Dr. Anthony Reinhart*

2013–2017 **B.S. Meteorology & B.A. in Mathematics**, University of Oklahoma
GPA: 4.0 / 4.0
Capstone: *The Effects of Veer-Back-Veer Wind Profiles on Storm Mode*

Research Publications

Journal Articles

1. **M. A. Satrio**, R. A. Saba, T. J. Pardun, J. C. Snyder, M. D. Flournoy, and M. C. Coniglio, "The Effects of a Veer–Back–Veer Wind Profile and the Effective Inflow-Layer Depth on Supercell Evolution," *Wea. Forecasting*, vol. 41, pp. 1689–1718, 2026. DOI: 10.1175/WAF-D-25-0240.1.
2. **M. A. Satrio** et al., "Radar-Observed Precipitation Enhancement Following UAS-Based Glaciogenic Cloud Seeding during the 2025–2026 Rainmaker Season," *Atmosphere*, 2026, In revision.
3. **M. A. Satrio**, M. C. Coniglio, E. N. Rasmussen, C. L. Ziegler, D. M. Stechman, and A. E. Reinhart, "Low-Level Mesocyclone Evolution of a Cyclic Tornadic Supercell Observed during TORUS on 17 May 2019," *Mon. Wea. Rev.*, vol. 153, pp. 909–937, 2025. DOI: 10.1175/MWR-D-24-0051.1.
4. M. Wagner et al., "Harnessing UAS and High-Resolution Satellite Imagery to Better Characterize Wind Damage and Understand Tornado Behavior," *Bull. Amer. Meteor. Soc.*, vol. 106, E492–E508, 2025. DOI: 10.1175/BAMS-D-23-0234.1.
5. M. E. Anderson et al., "Terrain Effects on the 13 April 2018 Mountainburg, Arkansas EF2 tornado.," *J. Operational Meteor.*, vol. 10, pp. 18–29, 2022. DOI: 10.15191/nwajom.2022.1002.
6. M. D. Flournoy, A. W. Lyza, **M. A. Satrio**, M. R. Diedrichsen, M. C. Coniglio, and S. Waugh, "A Climatology of Cell Mergers with Supercells and their Association with Mesocyclone Evolution," *Mon. Wea. Rev.*, vol. 150, pp. 451–461, 2022. DOI: 10.1175/MWR-D-21-0204.1.
7. **M. A. Satrio**, D. J. Bodine, A. E. Reinhart, T. Maruyama, and F. T. Lombardo, "Understanding how Complex Terrain Impacts Tornado Dynamics using a Suite of High-Resolution Numerical Simulations," *J. Atmos. Sci.*, vol. 4, pp. 3277–3300, 2020. DOI: 10.1175/JAS-D-19-0321.1.

Conference Proceedings (Oral or Poster Presentation Given)

1. **M. A. Satrio**, J. Aikins, A. Dixon, K. Suski, C. Wessel, and M. Decker, "Evolution of Radar Seeding Signatures: Insights from the 2025–2026 Rainmaker Season," in *25th Conf. on Planned and Inadvertent Weather Modification, AMS Summit*, Madison, WI, 2026. URL: https://ams.confex.com/ams/2026Summit/meetingapp.cgi/Paper/483299.
2. **M. A. Satrio**, R. A. Saba, T. J. Pardun, J. C. Snyder, M. D. Flournoy, and M. C. Coniglio, "The Effects of a Veer-Back-Veer Wind Profile and Effective Inflow Layer Depth on Supercell Evolution," in *32nd Conf. on Severe Local Storms, AMS Summit*, Madison, WI, 2026. URL: https://ams.confex.com/ams/2026Summit/meetingapp.cgi/Paper/483773.
3. **M. A. Satrio**, M. C. Coniglio, C. L. Ziegler, D. M. Stechman, and M. B. Wilson, "Observed Updraft Characteristics Using Multi-Doppler Analyses from the TORUS Field Campaign," in *41st International Conference on Radar Meteorology*, Toronto, ON, Canada, 2025.

4. **M. A. Satrio**, M. C. Coniglio, E. Rasmussen, C. L. Ziegler, D. M. Stechman, and A. E. Reinhart, "On the Low-Level Mesocyclone Evolution of the 17 May 2019 McCook / Farnam, Nebraska Tornadic Supercell Observed during TORUS," in *31st Conf. on Severe Local Storms*, Virginia Beach, VA, 2024.

5. **M. A. Satrio**, R. A. Saba, T. J. Pardun, and M. C. Coniglio, "The Effects of Backing Winds within the Effective Inflow Layer on Supercell and Tornado-Like Vortices Using CM1," in *31st Conf. on Severe Local Storms*, Virginia Beach, VA, 2024.

6. **M. A. Satrio**, D. M. Stechman, M. C. Coniglio, C. L. Ziegler, A. E. Reinhart, and C. C. Weiss, "A Multi-Doppler Analysis of the 23 May 2022 Morton, TX Tornadic Supercell Sampled during TORUS," in *31st Conf. on Severe Local Storms*, Virginia Beach, VA, 2024.

7. **M. A. Satrio**, M. C. Coniglio, E. N. Rasmussen, C. Z. Ziegler, and D. M. Stechman, "A Triple-Doppler Analysis of the 17 May 2019 McCook / Farnam, NE Tornadic Supercell," in *40th Conference on Radar Meteorology*, Minneapolis, MN, 2023.

8. D. J. Bodine et al., "Understanding Complex Terrain Effects on Tornado Dynamics Using Tree-Fall Observations and High-Resolution Simulations," in *30th Conf. on Severe Local Storms*, Santa Fe, NM, 2022.

9. **M. A. Satrio**, E. N. Rasmussen, M. C. Coniglio, C. Z. Ziegler, and D. M. Stechman, "Analysis of the 17 May 2019 McCook / Farnam, NE Tornadic Supercell Observed during TORUS through a Triple-Doppler Retrieval and Diabatic Lagrangian Analysis Approach," in *30th Conf. on Severe Local Storms*, Santa Fe, NM, 2022.

10. **M. A. Satrio**, D. J. Bodine, A. E. Reinhart, T. Maruyama, and F. T. Lombardo, "Understanding How Complex Terrain Impacts Tornado Dynamics Using a Suite of High-Resolution Numerical Simulations," in *Mesoscale Processes Across Scales, AMS 101st Annual Meeting*, New Orleans, LA (Virtual), 2021.

11. **M. A. Satrio**, E. N. Rasmussen, M. C. Coniglio, C. Z. Ziegler, and D. M. Stechman, "Multi-Doppler Analysis of the 17 May 2019 McCook / Farnam, NE Tornadic Supercell during TORUS," in *Student and Early Career Severe Local Storms Conference*, Virtual, 2021.

12. **M. A. Satrio**, D. J. Bodine, A. E. Reinhart, T. Maruyama, and F. T. Lombardo, "Understanding How Complex Terrain Impacts Tornado Dynamics Using a Suite of High-Resolution Numerical Simulations," in *Severe Local Storms Symposium, AMS 100th Annual Meeting*, Boston, TX, 2020.

13. **M. A. Satrio**, D. J. Bodine, A. E. Reinhart, and T. Maruyama, "The Effects of Translation and Surface Roughness on Tornado Structure and Flow," in *17th Annual Student Conference, AMS 98th Annual Meeting*, Austin, TX, 2018.

14. **M. A. Satrio**, D. J. Bodine, A. E. Reinhart, and T. Maruyama, "The Effects of Varying Surface Roughness, Translational Velocity and Swirl Ratio on an Idealized Tornado," in *29th Conf. on Severe Local Storms*, Stowe, VT, 2018. URL: `https://ams.confex.com/ams/29SLS/webprogram/Manuscript/Paper348604/ExtAbstract.pdf`.

Academic Achievements & Awards

2016–2017	School of Meteorology Senior Undergraduate Academic Achievement Award
2015–2016	School of Meteorology Junior Undergraduate Academic Achievement Award
2016	Study Abroad at Monash University in Australia
2015–2016	Rex Inman Scholarship Recipient
2014–2015	Kelvin and Lisa Droegemeier Endowed Scholarship for Excellence in Meteorology Recipient
2013–2017	School of Meteorology Scholarship Recipient
	University of Oklahoma Academic Excellence Award
	President's Honor Roll